

Nitender Kumar

Associate Director | AI Architect & GenAI Leader | Driving AI strategy, scalable AI platforms, and LLM/RAG systems with measurable business impact
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Professional Summary

AI/GenAI platform leader with 13+ years of experience shaping business-critical AI systems across enterprise risk, connected vehicle analytics, and intelligent decision platforms. Proven track record of driving AI platform strategy, architecture governance, and cross-functional delivery for cloud-native systems processing **5TB+/day across 500K+ connected vehicles**, delivering **sub-second analytics**, **30% cloud cost reduction**, and faster release cycles. Deep expertise across **LLMs, RAG, MLOps, Azure, and AWS**, with sustained success standardizing reusable platforms, lowering downstream change cost, and translating complex business problems into production-grade AI capabilities.

Core Competencies

- **AI Platform Architecture:** AI platform design, distributed systems, event-driven architecture, architecture governance, platform standardization
 - **AI/GenAI & MLOps:** LLMs, RAG, agentic workflows, NLP, machine learning, model lifecycle management, MLOps
 - **Data Architecture & Analytics:** Streaming pipelines, ETL/ELT, lakehouse and warehouse design, data quality, observability
 - **Cloud & Reliability:** Azure, AWS, serverless systems, CI/CD, performance and cost optimization
 - **Leadership & Strategy:** AI strategy, technical roadmap ownership, stakeholder management, cross-functional execution, mentoring
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Leadership Highlights

- Drove architecture strategy and delivery for AI and data initiatives across **India, UK, and US**, aligning product, risk, operations, and engineering stakeholders around reusable platform roadmaps and standardized patterns across programs.
 - Built and mentored teams of up to **12 engineers**, scaling delivery capacity while raising engineering quality through reusable patterns and stronger execution discipline.
 - Led platform optimization programs that reduced cloud spend by **25-30% (~\$36K annually)** without compromising analytics throughput or delivery velocity.
 - Established CI/CD, MLOps, and engineering standards across teams, reducing deployment failures by **50%** and improving release predictability.
 - Served as a principal architecture partner to business and technology leaders, defining solution direction for regulated, data-intensive, and AI-first platforms.
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Professional Experience

Associate Director / AI Architecture Lead – KPMG India

Dec 2025 – Present | Bangalore, India

TPRM (Third Party Risk Management) – AI-Driven Enterprise Risk Platform

- Lead architecture for a **cloud-native TPRM platform** that standardizes questionnaire workflows, document review, and compliance decisioning, reducing manual handoffs and establishing a reusable operating model for enterprise risk teams.
- Chose an **event-driven integration model** over synchronous point-to-point workflows due to long-running, callback-heavy risk processes, increasing resiliency and throughput across enterprise risk workflows.
- Designed **intelligent document processing pipelines** with **Azure AI Document Intelligence** to extract, classify, and index compliance evidence at scale, reducing manual review effort and accelerating assessment cycles.
- Used **Cosmos DB** for distributed workflow state instead of relational process tracking to absorb high-volume callbacks, scale horizontally, and preserve end-to-end audit traceability across workflows.
- Defined **architecture governance and security controls** across L1/L2/L3 flows, data privacy, prompt-injection defense, and multi-tenant isolation, establishing a reusable control baseline for regulated GenAI workloads across teams.

- Drove architecture decisions across global **risk, compliance, and operations** teams, converting regulatory requirements into reusable platform capabilities that accelerate rollout and reduce one-off solution design.

Architecture Copilot Studio – GenAI Architecture Intelligence Platform

- Lead **Architecture Copilot Studio**, a shared **architecture intelligence platform** that standardizes architecture workflows, reduces dependency on manual design reviews, and converts prompts and Mermaid definitions into editable **Draw.io** diagrams with persistent conversations and canvas-based editing.
- Designed **architecture-to-cloud mapping** and **scale-based cost estimation** at design time rather than downstream in delivery, enabling earlier feasibility decisions and lowering downstream rework across **AWS/Azure** programs.
- Established **reusable agentic workflows** and reference patterns for architecture generation, orchestration, and design optimization, giving multiple teams a shared architecture intelligence layer instead of bespoke tooling.

Principal Data Engineer – Darby Telematics India Pvt Ltd & Danlaw Inc

Mar 2019 – Nov 2025 | Bangalore, India

Deep View Analytics – Cloud-Native AI & Data Platform (Automotive Domain)

- Owned end-to-end architecture of **Deep View Analytics**, an **AWS-native AI and data platform** processing **5TB+/day across 500K+ connected vehicles**, serving as the business-critical shared backbone for vehicle health, driver risk, telematics, and emissions analytics.
- Chose **streaming-first ingestion and analytics** using **AWS Kinesis, Lambda, Glue, Redshift, and Athena** over batch-heavy processing because fleet operations required **sub-second** signals, accelerating intervention and decision cycles.
- Integrated **AWS SageMaker pipelines** for **predictive maintenance, driver risk scoring, and emissions forecasting**, establishing a repeatable ML lifecycle that moved high-value analytics use cases from experimentation into production.
- Used **serverless orchestration** for ingestion, enrichment, validation, and model inference instead of always-on schedulers and services, reducing operational overhead while running multiple AI and analytics workloads on a shared platform.
- Established **self-service analytics** and **GenAI-driven copilot experiences** on top of the platform, putting insights directly in the hands of operations and business teams through natural language workflows and accelerating decision cycles without engineering mediation.
- Owned platform decisions across ingestion, storage, warehousing, ML, and cost controls, standardizing a single backbone for operational monitoring, data science, and business reporting while reducing duplicated pipeline logic across teams.
- Led a team of **8 engineers** and institutionalized **CI/CD, data quality automation, and delivery standards**, reducing **release time by 40%** and **deployment failures by 50%**.
- Drove **data lifecycle, storage, and query optimization** across the platform, reducing cloud cost by **30%** while preserving analytics performance at production scale.

Senior Software Engineer – NEC India Pvt Ltd

Aug 2012 – Feb 2019 | Bangalore, India

- Designed **Hadoop-based data ingestion and processing pipelines** for multi-source enterprise datasets, standardizing validation and transformation flows to strengthen downstream data reliability.
- Developed a **computer vision-based Queue Length Management system** using **TensorFlow and OpenCV** to detect retail congestion in near real time, reducing customer wait times by **30%**.
- Architected a **Predictive Auto Ordering System** that translated demand signals into inventory recommendations, reducing stockouts by **20%** and improving sales continuity.
- Delivered NEC's **IoT Big Data Solution** through modular server and distributed analytics deployments, reducing client setup time by **50%** and accelerating rollout of data-driven applications.

Education

- **M.Tech, Data Science & Engineering** – BITS Pilani (2020–2022)
 - **B.Tech, Electronics & Communication** – GEIT, Dehradun (2007–2011)
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